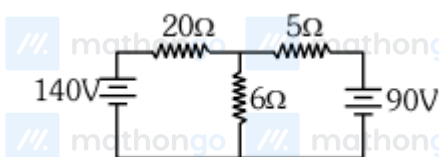


Q1 (20 July 2021 Shift 1)



The value of current in the 6Ω resistance is :

(1) 4 A

(2) 8 A

(3) 10 A

(4) 6 A

Q2 (20 July 2021 Shift 1)

A current of 5 A is passing through a non-linear magnesium wire of cross-section 0.04 m^2 . At every point the direction of current density is at an angle of 60° with the unit vector of area of cross-section. The magnitude of electric field at every point of the conductor is :

(Resistivity of magnesium $\rho = 44 \times 10^{-8} \Omega\text{m}$)

(1) $11 \times 10^{-2} \text{ V/m}$

(2) $11 \times 10^{-7} \text{ V/m}$

(3) $11 \times 10^{-5} \text{ V/m}$

(4) $11 \times 10^{-3} \text{ V/m}$

Q3 (20 July 2021 Shift 2)

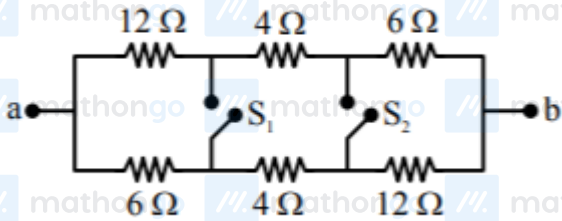
In the given figure switches S_1 and S_2 are in open

condition. The resistance across ab when the

Questions with Answer Keys

MathonGo

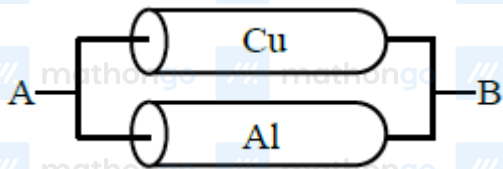
switches S_1 and S_2 are closed is _____ Ω



Q4 (22 July 2021 Shift 1)

A Copper (Cu) rod of length 25 cm and crosssectional area 3 mm^2 is joined with a similar Aluminium (Al) rod as shown in figure. Find the resistance of the combination between the ends A

and B. (Take Resistivity of Copper = $1.7 \times 10^{-8} \Omega \text{m}$ Resistivity of Aluminium = $2.6 \times 10^{-8} \Omega \text{m}$)



- (1) $2.170 \text{ m}\Omega$
- (2) $1.420 \text{ m}\Omega$
- (3) $0.0858 \text{ m}\Omega$
- (4) $0.858 \text{ m}\Omega$

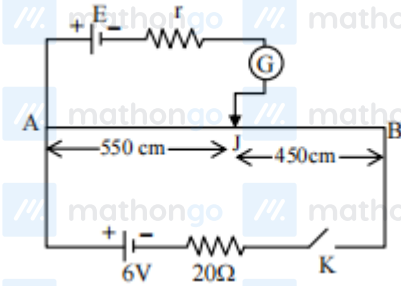
Q5 (22 July 2021 Shift 1)

In an electric circuit, a cell of certain emf provides a potential difference of 1.25 V across a load resistance of 5Ω . However, it provides a potential difference of 1 V across a load resistance of 2Ω

The emf of the cell is given by $\frac{x}{10} \text{ V}$. Then the value of x is _____.

Q6 (25 July 2021 Shift 1)

In the given figure, there is a circuit of potentiometer of length $AB = 10$ m. The resistance per unit length is 0.1Ω per cm. Across AB, battery of emf E and internal resistance ' r ' is connected. The maximum value of emf measured by this potentiometer is :



- (1) 5 V
- (2) 2.25 V
- (3) 6 V
- (4) 2.75 V

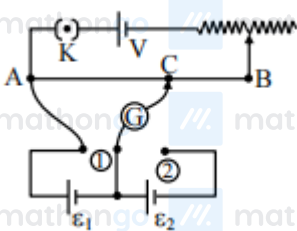
Q7 (25 July 2021 Shift 1)

An electric bulb rated as 200 W at 100 V is used in a circuit having 200 V supply. The resistance ' R ' that must be put in series with the bulb so that the bulb delivers the same power is Ω .

Q8 (25 July 2021 Shift 2)

In the given potentiometer circuit arrangement, the balancing length AC is measured to be 250 cm. When the galvanometer connection is shifted from point (1) to point (2) in the given diagram, the balancing length

becomes 400 cm. The ratio of the emf of two cells, $\frac{\varepsilon_1}{\varepsilon_2}$ is



- (1) $\frac{5}{3}$

Questions with Answer Keys

MathonGo

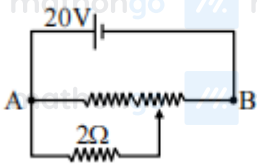
(2) $\frac{8}{5}$

(3) $\frac{4}{3}$

(4) $\frac{3}{2}$

Q9 (25 July 2021 Shift 2)

The given potentiometer has its wire of resistance 10Ω . When the sliding contact is in the middle of the potentiometer wire, the potential drop across 2Ω resistor is :



(1) 10 V

(2) 5 V

(3) $\frac{40}{9}$ V

(4) $\frac{40}{11}$ V

Q10 (25 July 2021 Shift 2)

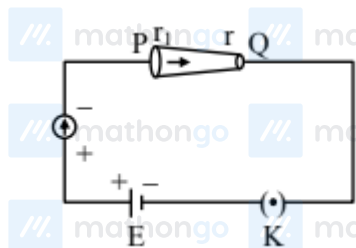
A 16Ω wire is bent to form a square loop. A 9 V supply having internal resistance of 1Ω is connected across one of its sides. The potential drop across the diagonals of the square loop is $\times 10^{-1}$ V

Q11 (27 July 2021 Shift 1)

In the given figure, a battery of emf E is connected across a conductor PQ of length ' l ' and different area of cross-sections having radii r_1 and r_2 ($r_2 < r_1$).

Questions with Answer Keys

MathonGo



Choose the correct option as one moves from P to Q :

- (1) Drift velocity of electron increases.
- (2) Electric field decreases.
- (3) Electron current decreases.
- (4) All of these

Q12 (27 July 2021 Shift 2)

The resistance of a conductor at 15°C is 16Ω and

at 100°C is 20Ω . What will be the temperature

coefficient of resistance of the conductor?

- (1) $0.010^{\circ}\text{C}^{-1}$
- (2) $0.033^{\circ}\text{C}^{-1}$
- (3) $0.003^{\circ}\text{C}^{-1}$
- (4) $0.042^{\circ}\text{C}^{-1}$

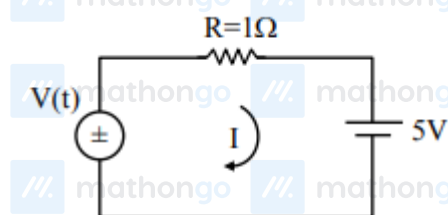
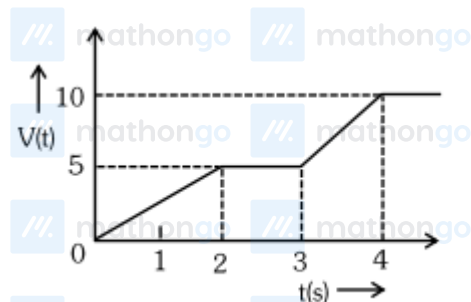
Q13 (27 July 2021 Shift 2)

For the circuit shown, the value of current at time

$t = 3.2\text{ s}$ will be A.

Questions with Answer Keys

MathonGo



[Voltage distribution $V(t)$ is shown by Fig. (1) and

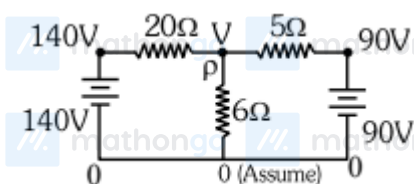
the circuit is shown in Fig. (2)]

Answer Key**Q1 (3)****Q2 (3)****Q3 (10)****Q4 (4)****Q5 (15)****Q6 (1)****Q7 (50)****Q8 (1)****Q9 (3)****Q10 (45)****Q11 (1)****Q12 (3)****Q13 (1)**

Hints and Solutions

MathonGo

Q1



Applying KCL at point P, $\frac{V-0}{6} + \frac{V-90}{5} + \frac{V-140}{20} = 0$

$$\Rightarrow 10V + 12V - 1080 + 3V - 420 = 0$$

$$\Rightarrow V = 60$$

$$\therefore \text{current in } 6\Omega = \frac{V-0}{6} = 10 \text{ A}$$

Hence option 3.

Q2

$$I = \vec{J} \cdot \vec{A} = JA \cos(\theta)$$

$$5 = J \left(\frac{4}{100} \right) \times \cos(60)$$

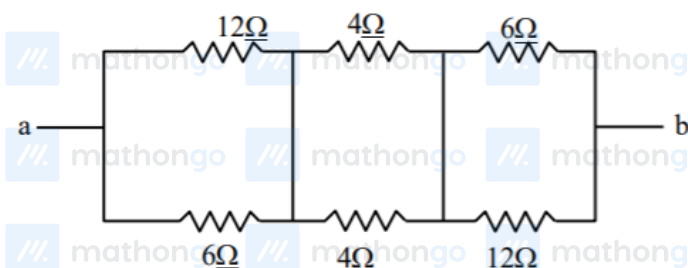
$$J = 5 \times 50 = 250 \text{ A/m}^2$$

$$\text{Now, } \vec{E} = \rho \cdot \vec{J}$$

$$= 44 \times 10^{-8} \times 250 = 11 \times 10^{-5} \text{ V/m}$$

Q3

When switch S_1 and S_2 are closed



$$\frac{12 \times 6}{12+6} + 2 + \frac{6 \times 12}{6+12}$$

$$\frac{72}{18} + 2 + \frac{72}{18} = 4 + 2 + 4 = 10\Omega$$

Hints and Solutions

MathonGo

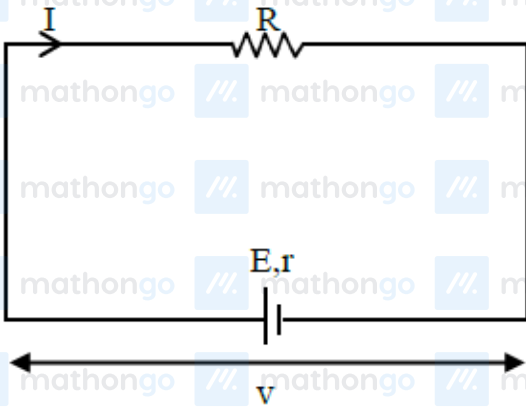
Q4

$$R = \frac{R_1 R_2}{R_1 + R_2} = \frac{\ell}{A} \cdot \frac{\rho_1 \rho_2}{\rho_1 + \rho_2}$$

$$R = \frac{25 \times 10^{-2}}{3 \times 10^{-6}} \times \frac{1.7 \times 2.6 \times 10^{-16}}{4.3 \times 10^{-8}}$$

$$R = 0.858 \text{ m}\Omega$$

Q5



$$\text{Terminal voltage } v = iR = \frac{ER}{R+r}$$

$$1^{\text{st}} \rightarrow 1.25 = \frac{E(5)}{5+r} \dots (i)$$

$$2^{\text{nd}} \rightarrow 1 = \frac{E(2)}{2+r} \dots (ii)$$

By (i) and (ii)

$$r = 1\Omega \quad E = \frac{3}{2} \text{ V} = \frac{15}{10} \text{ volt}$$

$$\Rightarrow x = 15$$

Q6

Max, voltage that can be measured by this potentiometer will be equal to potential drop across AB

$$R_{AB} = 10 \times 0.1 \times 100 = 100 \text{ ohm.}$$

$$V_{AB} = \frac{6}{20+100} \times 100 = 6 \times \frac{100}{120} = 5 \text{ V}$$

Q7

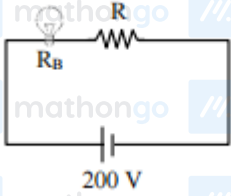
Hints and Solutions

MathonGo

$$\text{Power } P = \frac{V^2}{R_B}$$

$$R_B = \frac{V^2}{P} = \frac{100 \times 100}{200}$$

$$R_B = 50\Omega$$



To produce same power, same voltage (i.e. 100 V) should be across the bulb Hence, $R = R_B$

$$R = 50\Omega$$

Q8

$$E_1 = k\ell_1 \dots (i)$$

$$E_1 + E_2 = k\ell_2 \dots (ii)$$

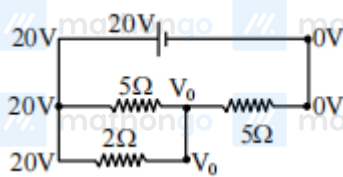
$$\frac{E_1}{E_1 + E_2} = \frac{\ell_1}{\ell_2} = \frac{250}{400} = \frac{5}{8}$$

$$8E_1 = 5E_1 + 5E_2$$

$$3E_1 = 5E_2$$

$$\frac{E_1}{E_2} = \frac{5}{3}$$

Q9



$$\frac{20 - V_0}{5} + \frac{0 - V_0}{5} + \frac{20 - V_0}{2} = 0$$

$$4 + 10 = \frac{2V_0}{5} + \frac{V_0}{2}$$

$$14 = \frac{4V_0 + 5V_0}{10}$$

$$V_0 = \frac{140}{9} \text{ Volt}$$

Hints and Solutions

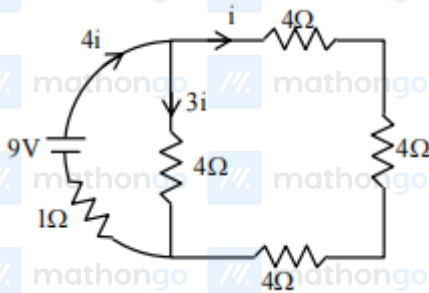
MathonGo

Potential difference across 2Ω resistor is $20 - V_0$ That is $\left(20 - \frac{140}{9}\right)$ Volt

Hence answer is $\left(\frac{40}{9}\right)$ Volt

Q10

here assume current as

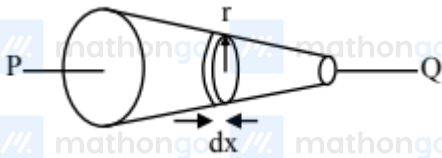


By KVL in outer loop $9 - 12i - 4i = 0$

$$8i = \frac{9}{2} = 4.5$$

$$= 45 \times 10^{-1}$$

Q11



Current is constant in conductor $i = \text{constant}$

$$\text{Resistance of element } dR = \frac{\rho dx}{\pi r^2}$$

$$dV = i dR = \frac{i \rho dx}{\pi r^2}$$

$$E = \frac{dV}{dx} = \frac{i \rho}{\pi r^2}$$

$$\& V_d = \frac{e E \tau}{m}$$

$$\therefore V_d \propto E$$

$$\rightarrow E \propto \frac{1}{r^2}$$

if r decreases, E will increase $\therefore V_d$ will increase

Hints and Solutions

MathonGo

Q12

$$16 = R_0 [1 + \alpha (15 - T_0)]$$

$$20 = R_0 [1 + \alpha (100 - T_0)]$$

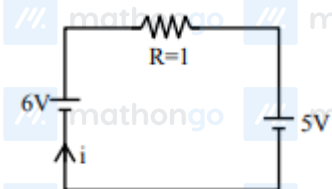
Assuming $T_0 = 0^\circ\text{C}$, as a general convention.

$$\Rightarrow \frac{16}{20} = \frac{1 + \alpha \times 15}{1 + \alpha \times 100}$$

$$\Rightarrow \alpha = 0.003^\circ\text{C}^{-1}$$

Q13

From graph voltage at $t = 3.2\text{sec}$ is 6 volt.



$$i = \frac{6-5}{1}$$

$$i = 1\text{ A}$$